

Nonproperness Components in a Degree-Six Keller Family

A human proof synchronized with the complete Lean 4 formalization
Including the universal family and infinite inequivalent subfamily

1 August 2026

Protocol A–F edition, final revision AF-5

Abstract

For the degree-six subfamily of the polynomial Keller maps in Theorem 5.2 of *A Counterexample to the Jacobian Conjecture*, we determine the reduced nonproperness set. If the only nonzero deformation parameter is $\eta_4 = h \in \mathbb{C}[T] \setminus \{0\}$, then

$$S_{F_h} = D_h \cup V(p^6 h(p)),$$

where D_h is one irreducible nonvertical hypersurface and contains no vertical hyperplane. Consequently

$$\#\mathrm{Irr}(S_{F_h}) = 2 + \#\{\alpha \in \mathbb{C}^\times : h(\alpha) = 0\},$$

with roots counted without multiplicity. This proves that CEX-004 and CEX-006 are not left-right polynomially equivalent and yields infinitely many pairwise inequivalent Keller counterexamples of generic degree six.

The accompanying Lean 4 project is pinned to Lean 4.32.2 and to the exact mathlib revision recorded in Section D. It now proves the family-wide Jacobian identity, the genuine function-field generic degree, the degree-one obstruction for polynomial automorphisms, the reduced nonproperness equality for every nonzero h , the universal component formula, and the explicit infinite pairwise inequivalent family. The parameter-free declarations `theoremA` and `theoremB` assemble these facts exactly.

The final pinned replay completed successfully with 3507 build jobs. Its enforced audit parsed 92 unique `#print axioms` reports and accepted only `propext`, `Classical.choice`, and `Quot.sound`; the project source scan found no project-local `axiom`, `sorry`, or `admit`. The release manifest and archive are verified separately. All nonproperness and zero-locus equalities are equalities of underlying reduced sets; no scheme-theoretic multiplicity statement is asserted.

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A Theorem intake

A.1 Exact objects, assumptions, and equality notion

For a polynomial map $G : \mathbb{C}^n \rightarrow \mathbb{C}^n$, write

$$S_G = \left\{ b \in \mathbb{C}^n : \begin{array}{l} \text{there exist } u_k \in \mathbb{C}^n \text{ with } \|u_k\| \rightarrow \infty, \\ \text{and } G(u_k) \rightarrow b \end{array} \right\}$$

for its nonproperness set. Throughout, S_G denotes its underlying reduced algebraic set. Thus every equality of nonproperness loci below is equality of sets, and every irreducible-component count ignores scheme multiplicities.

The base field is $\mathbb{C}$. In the family F_h we assume $h \in \mathbb{C}[T] \setminus \{0\}$. Left-right polynomial equivalence means

$$G = L \circ F \circ R, \quad L, R \in \text{Aut}_{\text{poly}}(\mathbb{C}^3).$$

The proof uses classical polynomial algebra, Euclidean topology on complex affine space, and standard algebraic geometry for constructible images, dimension, Zariski closure, and irreducibility.

The intermediate formal relation used by Lean is slightly more explicit. It requires Euclidean homeomorphisms L and R , preservation of escaping sequences by R , a target affine-Zariski homeomorphism L_z compatible with L on points, and the identity $G = L \circ F \circ R$. This relation is named `AlgebraicEscapeLeftRightEquivalent`. The completed `PolynomialAutomorphism` module represents polynomial forward and inverse coordinate triples with exact two-sided inverse laws, derives their Euclidean and affine-Zariski homeomorphisms, proves escape preservation, and shows that ordinary polynomial left-right equivalence supplies all data in the intermediate interface.

A.2 Claims and proof status

Item	Exact status	Level
Human mathematics	All displayed mathematical theorems in this document are proved from the stated hypotheses, with the standard geometric results identified where used.	H1
Universal Lean family theorem	For every nonzero h , Lean proves constant Jacobian -2 , function-field generic degree six, nonautomorphism, the exact reduced nonproperness equality, hypersurface and dimension-two finite geometry, and the actual component formula.	Checked, unconditional
Formal Theorem A	The parameter-free declaration <code>theoremA</code> proves the complete degree-six Keller-counterexample certificates, counts 3 and 2, and polynomial left-right inequivalence.	Checked, unconditional
Formal Theorem B	Lean proves the product family has count $m + 2$, an infinite range, and pairwise polynomial left-right inequivalence on that same range.	Checked, unconditional
Polynomial left-right bridge	Polynomial coordinate automorphisms with polynomial inverses induce the required Euclidean and affine-Zariski homeomorphisms and preserve component counts.	Checked, unconditional
Equality notion	Reduced algebraic sets only; no scheme-theoretic multiplicity is claimed.	Explicit
Trust boundary	No project-local <code>axiom</code> , <code>sorry</code> , or <code>admit</code> ; audited declarations use only <code>propext</code> , <code>Classical.choice</code> , and <code>Quot.sound</code> .	Audited

Theorem A. *The degree-six Keller counterexamples CEX-004 and CEX-006 are not left-right polynomially equivalent. Their reduced nonproperness sets have respectively three and two irreducible components:*

$$\#\text{Irr}(S_{\text{CEX-004}}) = 3, \quad \#\text{Irr}(S_{\text{CEX-006}}) = 2.$$

Here left-right polynomial equivalence means the notion fixed above.

Theorem B. *There are infinitely many pairwise left-right polynomially inequivalent Keller counterexamples $\mathbb{C}^3 \rightarrow \mathbb{C}^3$, all having generic degree six.*

Kernel-checked main result. *In the pinned Lean environment, for every nonzero $h \in \mathbb{C}[T]$,*

$$\text{NonpropernessSet}(F_h) = D_h \cup V(p^6 h(p)), \quad \#\text{Irr}(S_{F_h}) = 2 + \#\{\alpha \in \mathbb{C}^\times : h(\alpha) = 0\}.$$

The polynomial model has Jacobian determinant -2 , genuine function-field generic degree six, and no polynomial inverse. The parameter-free Lean `theoremA` gives the two explicit counts and their polynomial left-right inequivalence; `theoremB` gives the infinite, pairwise inequivalent product family with count $m + 2$.

The universal component formula is

$$\#\text{Irr}(S_{F_h}) = 2 + \#\{\alpha \in \mathbb{C}^\times : h(\alpha) = 0\},$$

where distinct roots are counted without multiplicity. The cases of constant h , a root at zero, repeated roots, and the empty product in the infinite family are included in the theorem as stated.

B Proof route

B.1 Shape and mathematical material

The central claim is an equality of reduced algebraic sets. The human proof uses double inclusion, followed by an irreducible-component count. It combines exact polynomial identities, a rational parametrization, elementary properness and covering arguments, and irreducibility of an image closure. The final polynomial inequivalence statement is obtained from invariance of the component count.

The completed Lean proof follows the universal route described by the human argument, with every topological transition made explicit. It reifies F_h as a polynomial coordinate map and computes the formal Jacobian determinant. It defines generic degree as an actual intermediate-field finrank and obtains degree six through the inverse generator and a rational-function field model. For nonproperness, it combines Keller local homeomorphisms, local properness, covering-space monodromy, dense good targets with exactly six simple inverse roots, and an `encard` bound of at most three on the leading locus. Exact eliminants and `mathlib`'s genuine `irreducibleComponents` then close the reduced-set equality and component formula. No numerical or external CAS result is imported as a hypothesis.

B.2 Proof obligations and their closure

1. reify the point map as `FhPolynomial` and prove its exact evaluation bridge;
2. compute the family-wide formal Jacobian determinant as the constant polynomial -2 ;
3. define function-field generic degree as a finrank, prove it is six for nonzero h , and prove polynomial automorphisms have degree one;
4. derive the inverse-root equation and reconstruction identities;
5. prove the finite critical image and its closure are irreducible, hypersurface, dimension two, dominant, and nonvertical;
6. prove local fiber constancy away from nonproperness using proper restricted Keller coverings and monodromy;
7. meet each such neighborhood with the dense good-target set and transport the exact six-point fiber count;
8. prove every leading-locus fiber is finite with `encard` at most three, forcing fiber loss and nonproperness;
9. combine this reverse inclusion with universal asymptotic exclusion and exact eliminants to obtain the reduced-set equality;
10. identify and count the genuine irreducible components, with repeated roots discarded and the zero root handled once; and

11. specialize to the explicit product family and transport component count through polynomial left-right equivalence.

Items 1–11 are kernel-checked for every nonzero deformation polynomial. The two parameter-free paper theorems are direct assemblies of these endpoints.

B.3 Representation and library audit

Euclidean complex affine space is represented by `C3 := Fin 3 -> Complex`. Its affine-Zariski topology is constructed on the separate type synonym `ZariskiC3`, preventing accidental mixing of analytic and algebraic topologies. The invariant `algebraicComponentCount` is defined as the cardinality of `mathlib`'s actual `irreducibleComponents` of the corresponding Zariski subspace.

The finite component is the Zariski closure of a structured critical image. The source of the critical map is the principal open $s \neq 0$ in affine 2-space. A reusable principal-open regular-map layer proves the critical map Zariski-continuous and its image irreducible. The universal eliminant identifies its closure with one nonzero-polynomial zero locus, proves nonverticality and dominance, and supports the exact topological Krull dimension computation. The analytic modules connect the same locus to the sequence-defined Euclidean nonproperness set. Section D records the exact module boundaries and endpoint declarations.

C Human proof

C.1 The degree-six subfamily

Set

$$A = 1 + xy, \quad B = A^2z + y^2(4 + 3xy), \quad P = AB,$$

and

$$Q = y + 3xB, \quad R = 2x - 3x^2y - x^3z.$$

Fix $h \in \mathbb{C}[T] \setminus \{0\}$. In Theorem 5.2 of [1], take $\eta_4 = h$ and $\eta_j = 0$ for $j \neq 4$. The resulting map is

$$F_h = (P, Q_h, R_h),$$

where

$$Q_h = Q + A^2x^4B^6h(P), \quad R_h = R - \frac{2}{3}x^6B^6h(P).$$

On the dense chart $A \neq 0$, put $s = x/A$. If $(p, q, r) = F_h(x, y, z)$, define

$$\phi_h(p, s) = 3ps + p^6h(p)s^4, \quad \theta_h(p, s) = ps^3 + \frac{2}{3}p^6h(p)s^6.$$

Then

$$q = y + \phi_h(p, s), \quad r = 2s - ys^2 - \theta_h(p, s).$$

Eliminating $y = q - \phi_h(p, s)$ gives

$$\boxed{\Omega_{p,q,r}(s) = a_h(p)s^6 + 2ps^3 - qs^2 + 2s - r = 0, \quad a_h(p) = \frac{1}{3}p^6h(p).} \quad (1)$$

Moreover,

$$\frac{\partial \Omega}{\partial s} = 2(1 - s(q - \phi_h(p, s))). \quad (2)$$

Writing

$$D = 1 - s(q - \phi_h(p, s)) = 1 - sy,$$

a finite root is simple exactly when $D \neq 0$. For such a root, the corresponding source point is

$$y = q - \phi_h(p, s), \quad x = \frac{s}{D}, \quad z = pD^3 - y^2(4 - sy)D. \quad (3)$$

Indeed, these expressions give $A = D^{-1}$ and $B = pD$, and hence $F_h(x, y, z) = (p, q, r)$.

Lemma C.1 (Keller property and generic degree). *For every nonzero h , the map F_h satisfies*

$$\det JF_h = -2, \quad \mu(F_h) = 6.$$

Proof. First observe that

$$\partial_s \theta_h = s^2 \partial_s \phi_h.$$

In the coordinates (P, y, s) , this gives

$$\det \frac{\partial(P, Q_h, R_h)}{\partial(P, y, s)} = 2 - 2ys - \partial_s \theta_h + s^2 \partial_s \phi_h = 2(1 - ys) = \frac{2}{A}.$$

On the other hand,

$$\det \frac{\partial(P, y, s)}{\partial(x, y, z)} = -A.$$

For completeness, $P_z = A^3$ and $s_x = A^{-2}$ with y fixed, so expansion along the y -row gives the displayed determinant. Thus $\det JF_h = -2$ on $A \neq 0$. Both sides are polynomial functions, so the identity holds on all of $\mathbb{C}^3$.

Now let $K = \mathbb{C}(p, q)$. Equation (1) has the form

$$f_{p,q}(s) - r = 0, \quad f_{p,q}(S) = a_h(p)S^6 + 2pS^3 - qS^2 + 2S.$$

Because $h \neq 0$, the element $a_h(p)$ is nonzero in K , so $f_{p,q}$ has degree six. The polynomial $T - f_{p,q}(S)$ generates a prime ideal of $K[S, T]$, since the quotient by that ideal is $K[S]$. It is therefore irreducible in $K[S, T]$, and Gauss's lemma makes $f_{p,q}(S) - r$ irreducible in $K(r)[S]$. Finally, (3) gives

$$\mathbb{C}(x, y, z) = \mathbb{C}(p, q, r)(s).$$

Consequently the induced function-field extension has degree six, so $\mu(F_h) = 6$. $\square$

C.2 Elementary facts about nonproperness

Lemma C.2 (Closedness). *For every polynomial map $G : \mathbb{C}^n \rightarrow \mathbb{C}^n$, the set S_G is closed in the Euclidean topology.*

Proof. Suppose $b \notin S_G$. Then there are $R > 0$ and $\varepsilon > 0$ such that

$$\|u\| > R \implies \|G(u) - b\| \geq \varepsilon.$$

Otherwise one could choose u_k with $\|u_k\| > k$ and $\|G(u_k) - b\| < 1/k$, contradicting $b \notin S_G$. Every point within $\varepsilon/2$ of b obeys the same conclusion with $\varepsilon/2$. Thus the complement of S_G is open. $\square$

Lemma C.3 (Fiber loss forces nonproperness). *Let $G : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be a Keller map of generic degree d . If $b \notin S_G$, then*

$$\#G^{-1}(b) = d.$$

Equivalently, any target whose fiber contains fewer than d points lies in S_G .

Proof. If $b \notin S_G$, the argument in Lemma C.2 gives a connected neighborhood U of b over which G is proper. Since G is Keller, it is a local biholomorphism. Its image in U is open, and properness makes that image closed. The neighborhood U intersects the Zariski-open dense set of targets having generic fiber cardinality d , so the image is nonempty. Connectedness of U therefore makes the map surjective. A proper surjective local biholomorphism is a finite covering map, so its fiber cardinality is constant on U . Since that cardinality equals d at a generic target, it equals d over b . $\square$

C.3 The finite-multiple-root component

Let D_h be the Zariski closure of the targets for which (1) has a finite multiple root. Since

$$\partial_s \Omega_{p,q,r}(0) = 2,$$

such a root is never zero. Solving $\Omega = \partial_s \Omega = 0$ for q and r gives

$$\boxed{\begin{aligned} \Psi_h : \mathbb{A}_p^1 \times \mathbb{G}_{m,s} &\longrightarrow \mathbb{A}_{p,q,r}^3, \\ (p, s) &\longmapsto \left(p, p^6 h(p) s^4 + 3ps + \frac{1}{s}, s - ps^3 - \frac{2}{3} p^6 h(p) s^6 \right). \end{aligned}} \quad (4)$$

Lemma C.4. *The variety D_h is an irreducible hypersurface. It dominates the p -line, is distinct from every vertical hyperplane $V(p - \alpha)$, and contains no vertical hyperplane.*

Proof. The domain $\mathbb{A}^1 \times \mathbb{G}_m$ of Ψ_h is irreducible, so the closure of its image is irreducible. Its dimension is two: the first target coordinate is p , and

$$\frac{\partial q}{\partial s} = 4p^6 h(p) s^3 + 3p - \frac{1}{s^2}$$

is not identically zero. Hence the differential of Ψ_h has rank two on a nonempty Zariski-open subset. By the characteristic-zero dimension theorem for images, the closure D_h has dimension two. It is therefore an irreducible surface in $\mathbb{A}^3$, hence an irreducible hypersurface.

The first coordinate of $\Psi_h(p, s)$ is the freely varying parameter p , so D_h dominates the p -line. A vertical hyperplane does not dominate the p -line, and therefore cannot equal D_h . Finally, if a vertical hyperplane $H_\alpha = V(p - \alpha)$ were contained in D_h , then the two irreducible closed sets H_α and D_h would have the same dimension (two). A proper inclusion of irreducible closed subsets strictly lowers dimension, so the inclusion would force $H_\alpha = D_h$, contradicting the preceding paragraph. Thus no vertical hyperplane is contained in D_h . $\square$

C.4 The reduced nonproperness theorem

Theorem C.5. *For every nonzero $h \in \mathbb{C}[T]$,*

$$\boxed{S_{F_h} = D_h \cup V(a_h) = D_h \cup V(p^6 h(p))} \quad (5)$$

as reduced algebraic sets.

Proof. We prove both inclusions.

First inclusion: $S_{F_h} \subseteq D_h \cup V(a_h)$. Take $b_0 = (p_0, q_0, r_0) \notin D_h \cup V(a_h)$. Then $a_h(p_0) \neq 0$, and Ω_{b_0} is a degree-six polynomial with six simple finite roots. Choose a sufficiently small compact target neighborhood K of b_0 disjoint from $D_h \cup V(a_h)$. On K , the leading coefficient $a_h(p)$ stays bounded away from zero. The coefficients of Ω remain bounded, so the usual polynomial root bound places every root in one fixed compact disk.

The incidence set

$$\mathcal{I}_K = \{(b, s) \in K \times \mathbb{C} : \Omega_b(s) = 0\}$$

is therefore compact. No point of $\mathcal{I}_K$ satisfies $\partial_s \Omega = 0$, because $K \cap D_h = \emptyset$. Hence, by (2), $|D|$ has a positive minimum on $\mathcal{I}_K$. Formula (3) then shows that all corresponding source points are uniformly bounded.

There are no additional source points on $A = 0$: that equation forces $P = 0$, whereas p stays away from zero on K . Consequently $F_h^{-1}(K)$ is compact, and $b_0 \notin S_{F_h}$. This proves the first inclusion.

Second inclusion, finite component: $D_h \subseteq S_{F_h}$. Let $(p, q, r) = \Psi_h(p, t)$, where $t \neq 0$. For $\varepsilon \neq 0$, set

$$y_\varepsilon = \frac{1 - \varepsilon}{t}, \quad x_\varepsilon = \frac{t}{\varepsilon}, \quad z_\varepsilon = p\varepsilon^3 - y_\varepsilon^2(4 - ty_\varepsilon)\varepsilon.$$

Then $A = \varepsilon^{-1}$ and $s = t$. Direct substitution gives

$$F_h(x_\varepsilon, y_\varepsilon, z_\varepsilon) = \left(p, q - \frac{\varepsilon}{t}, r + \varepsilon t\right) \longrightarrow (p, q, r).$$

At the same time, $|x_\varepsilon| = |t/\varepsilon| \rightarrow \infty$. Thus every point in the image of Ψ_h belongs to S_{F_h} . The image of Ψ_h is constructible and Zariski dense in D_h ; hence it contains a Zariski-open dense subset of D_h , which is also Euclidean dense. Lemma C.2 now gives $D_h \subseteq S_{F_h}$.

Second inclusion, vertical components: $V(a_h) \subseteq S_{F_h}$. The distinct roots of $a_h(p) = \frac{1}{3}p^6 h(p)$ are 0 together with the distinct nonzero roots of h . Fix one such root α , and consider a generic target on $H_\alpha = V(p - \alpha)$ outside D_h .

If $\alpha \neq 0$, then $h(\alpha) = 0$, and (1) specializes to

$$2\alpha s^3 - qs^2 + 2s - r = 0.$$

For a target outside D_h , this cubic has three simple roots. Equation (3) gives exactly three source points with $A \neq 0$. There are no source points with $A = 0$, because $A = 0$ forces $P = 0 \neq \alpha$. Hence the fiber has three points.

If $\alpha = 0$, then, for generic $q \neq 0$, equation (1) becomes

$$-qs^2 + 2s - r = 0.$$

Outside D_h , it has two simple finite roots and hence gives two source points with $A \neq 0$. There is exactly one further source point on $A = 0$. Indeed, $A = 0$ implies $xy = -1$, $B = y^2$, and $P = 0$. Moreover,

$$Q_h = -2y, \quad R_h = 5x - x^3z - \frac{2}{3}y^6h(0).$$

Thus, when $q \neq 0$,

$$y = -\frac{q}{2}, \quad x = \frac{2}{q}, \quad z = \frac{5x - \frac{2}{3}y^6h(0) - r}{x^3},$$

which is one and only one point. The generic fiber over H_0 therefore also has three points.

By Lemma C.1, F_h has generic degree six. Lemma C.3 now implies that every target in these generic open subsets of the vertical hyperplanes lies in S_{F_h} , because its fiber has only three points. Each such open subset is Euclidean dense in its hyperplane: Lemma C.4 ensures that D_h contains no vertical hyperplane. Since S_{F_h} is closed by Lemma C.2, every vertical hyperplane H_α is contained in S_{F_h} .

This proves $V(a_h) \subseteq S_{F_h}$ and completes (5). $\square$

C.5 Earlier pair-specific explicit asymptotic witnesses

An earlier accepted checkpoint established the two concrete vertical loci by explicit escaping inverse-root branches. Those witnesses remain useful independent checks, but the final universal theorem uses the fiber-loss route: local proper Keller coverings have locally constant finite fibers, dense good targets have exactly six simple inverse roots, and every leading-locus fiber has cardinality at most three. The explicit pair witnesses are recorded here because they give a second proof for the two original examples.

For the common component $p = 0$, let $\varepsilon \rightarrow 0$, put

$$s_\varepsilon = \varepsilon^{-5}, \quad d(q) = -\frac{q}{10}, \quad w_\varepsilon = c + d(q)\varepsilon^2, \quad p_\varepsilon = \varepsilon^3 w_\varepsilon,$$

and choose

$$c^5 = -6 \quad \text{for CEX-004}, \quad c^5 = 4 \quad \text{for CEX-006}.$$

For a prescribed target $(0, q, r)$ define

$$q_\varepsilon = a_h(p_\varepsilon)s_\varepsilon^4 + 2p_\varepsilon s_\varepsilon + \frac{2}{s_\varepsilon} - \frac{r}{s_\varepsilon^2}.$$

Then s_ε is an exact root of $\Omega_{p_\varepsilon, q_\varepsilon, r}$. The cancellation conditions on c give

$$p_\varepsilon \longrightarrow 0, \quad q_\varepsilon \longrightarrow q.$$

The evaluated derivative is eventually nonzero, so the checked reconstruction map produces exact source points mapping to $(p_\varepsilon, q_\varepsilon, r)$. Moreover the reconstructed second source coordinate satisfies

$$y_\varepsilon \varepsilon^2 \longrightarrow 3c \neq 0,$$

and hence the sources escape. This proves the entire hyperplane $p = 0$ is contained in each pair-specific nonproperness set. It corrects the tempting but insufficient direct $A = 0$ family, which approaches only the line $(p, q) = (0, 0)$.

For the additional CEX-004 hyperplane, set $\alpha = -\frac{1}{4}$ and let $s \rightarrow \infty$. For a prescribed target (α, q, r) define

$$p_s = \alpha + \frac{1536}{s^3} + \frac{3072q}{s^4}, \quad q_s = a_{004}(p_s)s^4 + 2p_s s + \frac{2}{s} - \frac{r}{s^2}.$$

Again s is an exact root of $\Omega_{p_s, q_s, r}$, and the checked limits are

$$p_s \longrightarrow -\frac{1}{4}, \quad q_s \longrightarrow q, \quad s^{-2}\Omega'_{p_s, q_s, r}(s) \longrightarrow \frac{3}{2}, \quad s^{-1}y_s \longrightarrow -\frac{3}{4}.$$

Thus the root is eventually simple, reconstruction is exact, and the source sequence escapes. Consequently the whole hyperplane $p = -\frac{1}{4}$ lies in S_{004} .

The formal forward inclusion uses the inverse chart rather than a compact neighborhood argument. If $F_h(u_n) \rightarrow b$ and $a_h(b_0) \neq 0$, then eventually $A(u_n) \neq 0$ and the chart roots $s_n = x_n/A(u_n)$ satisfy the inverse equation. A direct Cauchy-type estimate bounds s_n ; after passage to a convergent subsequence, a nonzero limiting derivative would make the reconstructed sources converge, contradicting escape. The limiting root is therefore multiple and nonzero, so b lies in the finite critical image, hence in D_h . Lean proves the universal inclusion

$$\text{NonpropernessSet}(F_h) \subseteq D_h \cup V(a_h)$$

for every deformation polynomial h .

To pass from the directly parametrized critical image to its Zariski closure for the two examples, Lean does not infer Euclidean closure from Zariski closure. Instead it verifies exact elimination certificates. Two Euclidean remainder identities reduce Ω and Ω' to a cubic H_3 and a quadratic H_2 , and the fixed 5×5 Sylvester determinant gives

$$\text{Res}(H_3, H_2) = 27p^{10}\delta_{004}(p, q, r)$$

for CEX-004 and

$$\text{Res}(H_3, H_2) = -\frac{243}{8}p^{10}\delta_{006}(p, q, r)$$

for CEX-006. Off the leading locus, vanishing of the pair-specific factor produces a common finite root of Ω and Ω' , hence a point of the critical image. On the leading locus, the explicit vertical branches above already give nonproperness. This closes the full reverse inclusions $D_{004} \subseteq S_{004}$ and $D_{006} \subseteq S_{006}$ inside Lean.

C.6 Component count

Corollary C.6 (General component formula). *For every nonzero $h \in \mathbb{C}[T]$,*

$$\boxed{\#\text{Irr}(S_{F_h}) = 2 + \#\{\alpha \in \mathbb{C}^\times : h(\alpha) = 0\}.} \tag{6}$$

The roots are counted without multiplicity.

Proof. As a reduced set,

$$V(p^6 h(p)) = V(p) \cup \bigcup_{\substack{\alpha \in \mathbb{C}^\times \\ h(\alpha) = 0}} V(p - \alpha).$$

These vertical hyperplanes are irreducible and pairwise distinct. Lemma C.4 says that D_h is irreducible, is not contained in any vertical hyperplane, and contains no vertical hyperplane. Hence the displayed finite union is irredundant. Theorem C.5 therefore decomposes S_{F_h} into exactly one finite-multiple-root component, one component $V(p)$, and one component for every distinct nonzero root of h . $\square$

C.7 CEX-004 versus CEX-006

For CEX-004,

$$h_{004}(T) = 1 + 4T.$$

It has the single nonzero root $-\frac{1}{4}$. Hence

$$S_{004} = D_{004} \cup V(p) \cup V(1 + 4p), \quad \# \text{Irr}(S_{004}) = 3.$$

For CEX-006,

$$h_{006}(T) = -\frac{3}{2}.$$

It has no roots. Hence

$$S_{006} = D_{006} \cup V(p), \quad \# \text{Irr}(S_{006}) = 2.$$

Lemma C.7 (Left-right invariance). *If $G = L \circ F \circ R$, where L and R are polynomial automorphisms of $\mathbb{C}^n$, then*

$$S_G = L(S_F).$$

Proof. A polynomial automorphism is proper because its inverse is continuous and maps compact sets to compact sets. Consequently a sequence u_k escapes to infinity if and only if $R(u_k)$ escapes to infinity. Also,

$$G(u_k) \rightarrow b \iff F(R(u_k)) \rightarrow L^{-1}(b).$$

The defining sequence condition for nonproperness therefore gives $b \in S_G$ if and only if $L^{-1}(b) \in S_F$. $\square$

An algebraic automorphism maps irreducible components bijectively. Thus left-right equivalent maps have nonproperness sets with the same number of irreducible components. The counts 3 and 2 are different, so CEX-004 and CEX-006 are not left-right polynomially equivalent. This proves Theorem A.

C.8 The infinite family

For $m \geq 0$, define

$$h_m(T) = \prod_{i=1}^m (T - i),$$

where the empty product h_0 is 1. The polynomial h_m has exactly m distinct nonzero roots. Lemma C.1 gives

$$\det JF_{h_m} = -2, \quad \mu(F_{h_m}) = 6.$$

In particular, F_{h_m} is a Keller map but not a polynomial automorphism, because an automorphism has generic degree one. Thus every F_{h_m} is a Keller counterexample. Corollary C.6 gives

$$\# \text{Irr}(S_{F_{h_m}}) = m + 2.$$

If $m \neq n$, these component counts differ, so Lemma C.7 rules out a left-right polynomial equivalence between F_{h_m} and F_{h_n} . The family $\{F_{h_m}\}_{m \geq 0}$ is therefore infinite, pairwise inequivalent, and has constant generic degree six. This proves Theorem B.

D Lean 4 formalization

D.1 Pinned environment and reproducible audit

The project is pinned by

```
leanprover/lean4:v4.32.2
```

and mathlib commit

```
905b95818eb32af7874a58b427f50c1711a5e96c.
```

The PowerShell entry point `build_and_audit.ps1` verifies the exact toolchain pin, resolves that pinned Lake executable directly, builds the umbrella and axiom-audit targets, scans the project Lean sources for `axiom`, `sorry`, or `admit`, parses every `#print axioms` report, and fails on missing, duplicate, unexpected, or non-whitelisted results. It performs no dependency update or cache download.

The immutable CP-07 replay reported

```
Build completed successfully (3507 jobs).
AXIOM_WHITELIST_PASSED (92 reports)
BUILD_AND_AUDIT_PASSED
```

Every audited endpoint reported only

```
propext,    Classical.choice,    Quot.sound.
```

The CP-07 formal-input manifest contains 53 verified entries. After the final paper and documentation synchronization, the release manifest is regenerated and reverified before packaging.

D.2 Module map

The root module has 32 direct runtime imports. There are 48 submodules under `DegreeSixKeller/`: 33 non-Spike modules and 15 Spike-named modules. The separate `AxiomAudit` imports the umbrella and is compiled as a second target. Two historically named support modules, `FunctionFieldSpike` and `GenericDegreeGeneratorSpike`, remain live dependencies of the generic-degree layer; the other 13 Spike modules are outside the production closure and retained as proof-development evidence.

Module	Kernel-checked content
FetaCore, InverseChart	Point maps, inverse polynomial, exact chart, derivative, reconstruction, and leading-coefficient locus.
PolynomialModel, JacobianKeller	Polynomial coordinate triples, exact evaluation bridge, formal Jacobian, determinant -2 , and Keller certificate.
FunctionFieldSpike, GenericDegreeGenerator- Spike, GenericDegree	Source and coordinate function fields, inverse generator, rational-function transport, and genuine finrank-six theorem.
PolynomialAutomorphism, GenericDegreeAutomorphism	Polynomial inverse data, left-right bridge, automorphism degree one, and nonautomorphism of every nonzero family member.
AffineZariski, ComponentCounting	Affine-Zariski topology and exact component counting through mathlib's <code>irreducibleComponents</code> .
CriticalImageGeometry, GeneralFiniteGeometry, FiniteComponentDimension	Universal critical image, exact eliminant zero locus, irreducibility, hypersurface, dominance, nonverticality, and topological Krull dimension two.
KellerLocalHomeomorph, FiberLoss	Keller local homeomorphisms, compact local restrictions, proper coverings, monodromy, and locally constant fiber size.
GenericFiberSix, GoodTargetDensity LeadingFiberBounds	Exact six-point fibers on a dense set of simple-root targets. Finiteness and <code>encard</code> at most three on the complete leading locus, including $p = 0$.
AsymptoticExclusion, GeneralNonproperness GeneralComponents	Universal forward inclusion, fiber-loss reverse inclusion, and exact reduced nonproperness equality. Distinct nonzero-root indexing, edge cases, exact irreducible-component Finsets, and the formula $2 + \#\{\alpha \neq 0 : h(\alpha) = 0\}$.
InfiniteFamily	Product parameters, exact roots and counts, Keller/degree-six certificates, infinite range, and pairwise polynomial inequivalence.
Pair-specific legacy modules	Independent explicit escape/resultant proofs for CEX-004 and CEX-006 and their original component-count contradiction.
MainTheorems AxiomAudit	Parameter-free Lean declarations <code>theoremA</code> and <code>theoremB</code> . The 92 public <code>#print axioms</code> probes used by the enforced trust audit.

D.3 Exact checked endpoint declarations

Representative completed declarations are:

```
theorem Fh_jacobianDet (h : Complex[X]) :
  jacobianDet (FhPolynomial h) = MvPolynomial.C (-2)

theorem Fh_genericDegree_six (h : Complex[X]) (hh : Not (h = 0)) :
```

```

genericDegree (FhPolynomial h) = 6

theorem Fh_reducedNonpropernessEquality
  (h : Complex[X]) (hh : Not (h = 0)) :
  NonpropernessSet (Fh h) = generalReducedCandidate h

theorem Fh_componentCount (h : Complex[X]) (hh : Not (h = 0)) :
  algebraicComponentCount (NonpropernessSet (Fh h)) =
    2 + (nonzeroRoots h).card

theorem hFamily_componentCount (m : Nat) :
  algebraicComponentCount
    (NonpropernessSet (Fh (hFamily m))) = m + 2

#check DegreeSixKeller.theoremA
#check DegreeSixKeller.theoremB

```

Here `generalReducedCandidate h` expands to the finite component union the zero locus of $p^6h(p)$. The pair specializations recover

$$\text{NonpropernessSet}(F_{004}) = \text{finiteComponent}(\eta_{004}) \cup \text{pHyperplane}(0) \cup \text{pHyperplane}\left(-\frac{1}{4}\right)$$

and

$$\text{NonpropernessSet}(F_{006}) = \text{finiteComponent}(\eta_{006}) \cup \text{pHyperplane}(0).$$

The declaration `theoremA` additionally packages the two Keller, generic-degree-six, and nonautomorphism certificates. The declaration `theoremB` quantifies over every m , packages count $m + 2$, and proves both literal infinitude of the range and pairwise polynomial left-right inequivalence on that range. Neither declaration has a theorem parameter or an unproved geometric hypothesis.

D.4 From pair closure to the universal theorem

The two irreducibility obligations are closed by treating the critical parametrization as a regular map on the principal open $D(s)$ in affine 2-space. That domain is a nonempty open subset of irreducible affine space; its continuous image is irreducible, and the closure remains irreducible.

The two nonverticality obligations are closed by the exact elimination factors δ_{004} and δ_{006} . Each finite component lies in the corresponding zero locus. On every slice $p = \alpha$, $r = 0$, the resulting univariate polynomial in q has a fixed nonzero quadratic coefficient: -243 for CEX-004 and 216 for CEX-006. Hence every vertical hyperplane contains a point outside that zero locus and cannot lie inside the finite component.

The earlier two reduced-set equalities are closed by combining:

1. explicit escaping sequences for the critical image and all pair-specific vertical components;
2. the exact reduced-resultant bridge from the finite Zariski closure back to a common inverse root off the leading locus; and
3. the asymptotic-exclusion theorem proving that no other escaping limit is possible.

This is a direct formal proof of both inclusions, not a theorem parameter or an imported geometric certificate.

The final family-wide reverse inclusion uses the accepted fiber-loss argument, not an unavailable blanket identification of arbitrary fibers with function-field finrank. Away from the nonproperness set, local properness and the Keller local-homeomorphism theorem produce a finite covering over a target ball. Monodromy makes its fiber cardinality constant. Density supplies a good target in that ball with six simple inverse roots, so the original fiber also has six points. On the leading locus, the algebraic analysis proves finiteness and **encard** at most three; hence those targets must be nonproper. This covers $p = 0$, nonzero roots, repeated roots, and constants.

D.5 Human-to-Lean correspondence

Human proof step	Formal status
Coordinate definitions and inverse polynomial	Checked exactly.
Polynomial coordinate model and exact evaluation bridge	Checked for every h .
Jacobian determinant -2	Checked for every h .
Generic degree six	Checked for every nonzero h using a genuine function-field <code>finrank</code> .
Polynomial automorphisms have generic degree one	Checked; consequently every nonzero family member is nonautomorphic.
Closedness of S_G	Checked for the sequence-defined <code>NonpropernessSet</code> .
Fiber-loss theorem	Checked through local proper Keller coverings, monodromy, dense good targets, and exact six-point fibers.
Inverse chart, derivative, and reconstruction	Checked exactly, including exact mapping from every finite simple root and reconstruction of every source point on $A \neq 0$.
Critical-image irreducibility and closure irreducibility	Checked for every deformation polynomial h .
Critical image contained in nonproperness	Checked by the explicit ε -escape family.
Complete leading locus	Checked by finite fiber bounds with <code>encard</code> at most three, including the entire $p = 0$ hyperplane.
$S_{F_h} \subseteq D_h \cup V(a_h)$	Checked universally for every h by root bounds, subsequence extraction, and reconstruction contradiction.
$D_h \cup V(a_h) \subseteq S_{F_h}$	Checked for every nonzero h by the fiber-loss route.
Finite-component geometry	Closed, irreducible, exact hypersurface, topological Krull dimension two, dominant, and contains no vertical hyperplane.
Universal irreducible-component formula	Checked using actual <code>irreducibleComponents</code> .
Left-right transport of nonproperness	Checked for homeomorphisms with explicit escape preservation.
Component-count invariance	Checked for a target map equipped with a compatible affine-Zariski homeomorphism.
Polynomial automorphisms supply those data	Checked through polynomial forward/inverse coordinate data and the induced Euclidean and affine-Zariski homeomorphisms.
Final $3 \neq 2$ contradiction in the formal interface	Checked unconditionally.
Explicit product family and count $m + 2$	Checked, including the empty product, exact distinct roots, infinite range, and pairwise inequivalence.
Parameter-free Theorems A and B	Checked unconditionally.

D.6 Axiom, hole, and integrity audit

The audited public declarations report only

`propext,      Classical.choice,      Quot.sound.`

These are ordinary Lean/mathlib foundational dependencies, not project-specific mathematical assumptions. The use of `Classical.choice` is visible in constructions such as chosen fifth roots, chosen principal-open pullback data, and noncomputable component counts. The source scan found no project-local `axiom`, `sorry`, or `admit`. No CAS output, floating-point computation, or numerical resultant is trusted: the polynomial and determinant identities are elaborated and checked by Lean.

The final `SOURCE_MANIFEST.sha256` is generated after the paper and documentation edits and verified entry by entry before packaging. This integrity replay is independent of the logical kernel audit.

D.7 Exact remaining formal boundary

The mathematical content of Theorems A and B is closed in Lean. The formal equalities concern underlying reduced sets, and the component invariant is the cardinality of mathlib’s actual irreducible-component set. The only stated boundary is scheme-theoretic: multiplicities, valuations, normalizations, and scheme structures on the nonproperness locus are not defined or claimed.

E Dual closure audit

E.1 Mathematical closure audit: passed at H1

The human argument closes every stated mathematical obligation:

- The exact reduced-set equality, component formula, present-pair polynomial inequivalence, and infinite degree-six theorem are proved.
- The hypothesis $h \neq 0$ is used exactly where needed to keep the generic degree equal to six; distinct roots are counted without multiplicity.
- Constant h , a root at 0, repeated roots, the special component $p = 0$, and the empty product $m = 0$ are covered.
- Nonproperness is established independently of component counting, so the decomposition and the invariant are not used circularly.
- Both possible containments between the finite component and a vertical hyperplane are excluded.
- Equality always means equality of underlying reduced algebraic sets. No scheme structure is tacitly imported.

The completed Lean artifact independently checks the entire universal spine: polynomial and point-map agreement, Jacobian and degree, nonautomorphism, inverse formulas, finite geometry, proper-covering fiber loss, the reduced-set equality, actual component formula, and both parameter-free main theorems.

E.2 Formal closure audit: Theorems A and B passed

The parameter-free `theoremA` and `theoremB`, together with all their public supporting endpoints, are accepted by the pinned kernel. The 3,507-job replay and 92-report enforced axiom audit supply the immutable formal-closure evidence.

F Remaining frontier

There is no remaining theorem-level action item for the claims in this paper. An optional future project could define and study a scheme-theoretic or valuation-theoretic refinement of the nonproperness locus. Such a refinement would be a new theorem with new invariance obligations; factors and multiplicities in a raw resultant are not promoted to invariants here.

References

- [1] *A Counterexample to the Jacobian Conjecture*, working manuscript, 20 July 2026, Theorem 5.2.